Distance between a plane and point

Key Points

Distance is perpendicular to plane so use normal.

Line from point to plane is

$$\boldsymbol{r} = \boldsymbol{p} + t\boldsymbol{n}$$

Then solve intersection with plane to find the point on plane $\boldsymbol{b}$ then the distance is $|\boldsymbol{p} - \boldsymbol{b}|$

When the plane is given by $\boldsymbol{r} \cdot \boldsymbol{n} = d$ this is equivalent to

$$\frac{|\boldsymbol{p} \cdot \boldsymbol{n} - d|}{|\boldsymbol{n}|}$$

Basic Skills

Q1. For the following planes state the normal

a) $6x - 3y + z = -2$ **b)** $z = 5 + x - 3y$

Q2. Find the distance between the point (5,4,2) and (-4,-2,0)

Q3. Convert the vector form of a plane $\left(\boldsymbol{r} - \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}\right) \cdot \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} = 0$ into cartesian form finding the plane constant d

Q4. Find the point at which the line $\boldsymbol{r} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ intersects with the plane $5x - y + z = 33$

Competency Skills

Q5. For the plane $x + y + z = 9$ and the point (3,4,5)

a) Write the equation of the line going through the point in the direction of the normal to the plane

b) Find out where you line from part a) intersects the plane

c) Hence work out the distance between the point and plane

d) Check your answer by using the formula $\frac{|\boldsymbol{p} \cdot \boldsymbol{n} - d|}{|\boldsymbol{n}|}$

Q6. Find the distance between the plane $2x - y + 3z = 12$ and the origin

Q7. Find the distance between the point (1,2,1) and the plane with normal parallel to the z-axis going through the point (5,4,2)

Advanced Skills

Q8. For the line $\boldsymbol{r} = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and plane $\left(\boldsymbol{r} - \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}\right) \cdot \begin{pmatrix} -3 \\ 3 \\ k+2 \end{pmatrix} = 0$

a) Determine the values for k such that the plane and line do not intersect

b) When $k = 1$ explain why the line and plane are parallel

c) Show that the point (5,-4,12) is on the line

d) For $k = 1$ and by using the point (5,-4,12) determine the distance between the line and point

e) For $k = 1$ and by using the point (-1,2,0) confirm that the distance between the line and point is the same

Extension

Q9. Consider how we could use this method of a point and plane for skew lines

Hint: Consider what direction is perpendicular to both lines